

Progress Report

Nominal PASS Placement and Worst-Case Array Gain Under Bounded Position Errors

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1 Background and Objective

Pinching-Antenna Systems (PASS) achieve high array gain by combining spherical-wave propagation from a dielectric waveguide to a user with a waveguide-induced linear phase profile across the pinching antennas (PAs), enabling near-field focusing. However, the physical pinching points are susceptible to manufacturing/actuation errors, which perturb PA locations and destroy phase coherence.

Goal. This report completes the full analysis pipeline:

- 1) Derive the PASS baseband model and the array-gain expression $a(\Delta)$ from first principles;
- 2) Formalize the nominal placement as

$$\Delta^* \in \arg \max_{\Delta \in \mathcal{X}} a(\Delta), \quad a^* = \max_{\Delta \in \mathcal{X}} a(\Delta); \quad (1)$$

- 3) Construct a near-optimal placement Δ^* using the PASS phase-alignment condition

$$R(\Delta) + n_{\text{eff}}\Delta = C' + m\lambda, \quad m \in \mathbb{Z}, \quad (2)$$

where each Δ is solvable by Newton's method (or in closed form);

- 4) Under bounded position errors $|\delta_n| \leq \varepsilon$, derive analytical lower and upper bounds on the worst-case gain at Δ^* and outline numerical validation.

2 PASS Baseband Signal Model and Array Gain

2.1 Geometry and coordinate transformation

Following the PASS geometry:

- The waveguide is parallel to the x -axis at height d .
- A single-antenna user is located at $\mathbf{u} = [x_u, 0, 0]^T$.
- The n -th PA is at $\boldsymbol{\psi}_n = [x_n, 0, d]^T$, where $n \in \mathcal{N} \triangleq \{\pm 1, \dots, \pm N/2\}$.
- The feed point is $\boldsymbol{\psi}_0 = [x_0, 0, d]^T$.

Define the relative displacement

$$\Delta_n \triangleq x_n - x_u, \quad (3)$$

and the free-space distance

$$R_n(\Delta_n) \triangleq \|\mathbf{u} - \boldsymbol{\psi}_n\| = \sqrt{d^2 + \Delta_n^2}. \quad (4)$$

2.2 Composite PASS channel

The effective channel includes both free-space propagation and waveguide-induced phase:

$$h_n(\Delta_n) = \underbrace{\frac{\eta^{1/2}}{R_n(\Delta_n)} e^{-ik_0 R_n(\Delta_n)}}_{\text{free-space}} \cdot \underbrace{e^{-ik_0 n_{\text{eff}}(x_n - x_0)}}_{\text{waveguide}}, \quad (5)$$

where $\eta = \frac{c^2}{16\pi^2 f_c^2}$, $k_0 = \frac{2\pi}{\lambda}$, and n_{eff} is the effective refractive index (typical values 1.44–1.99).

2.3 Received signal and array gain

Assume total transmit power P is uniformly allocated across N PAs. The received baseband signal is

$$y = \sqrt{\frac{P}{N}} \sum_{n \in \mathcal{N}} h_n(\Delta_n) s + z, \quad (6)$$

where $z \sim \mathcal{CN}(0, \sigma^2)$. Define the normalized array gain as

$$a(\Delta) \triangleq \frac{\text{SNR}}{P/\sigma^2} = \frac{\eta}{N} \left| \sum_{n \in \mathcal{N}} \frac{e^{-i\Phi_n(\Delta_n)}}{R_n(\Delta_n)} \right|^2. \quad (7)$$

For clarity, we adopt the common simplification $x_0 = x_u$ (equivalently, absorb $x_u - x_0$ into a global phase constant). Then the total phase becomes

$$\Phi_n(\Delta_n) = k_0 R_n(\Delta_n) + k_0 n_{\text{eff}} \Delta_n = k_0 \left(\sqrt{d^2 + \Delta_n^2} + n_{\text{eff}} \Delta_n \right). \quad (8)$$

Key difference vs. conventional MA arrays. In MA-type free-space-only models, the phase depends mainly on $k_0 R_n(\Delta_n)$. PASS adds $k_0 n_{\text{eff}} \Delta_n$, and $n_{\text{eff}} > 1$ amplifies the phase sensitivity to position errors.

3 Nominal Optimized Placement via Phase Alignment

3.1 Optimization problem

We formalize nominal placement optimization as

$$\Delta^* \in \arg \max_{\Delta \in \mathcal{X}} a(\Delta), \quad (9)$$

where $\mathcal{X}$ is the feasible set, e.g., minimum spacing constraints

$$|\Delta_n - \Delta_{n'}| \geq \Delta_p \lambda, \quad \Delta_p \geq \frac{1}{2}, \quad (10)$$

and the common symmetry for a single user:

$$\Delta_{-n} = -\Delta_n, \quad n = 1, \dots, N/2. \quad (11)$$

3.2 Phase alignment condition

To approach the triangle-inequality upper envelope

$$\left| \sum_n c_n e^{-i\theta_n} \right| \leq \sum_n |c_n|, \quad (12)$$

we enforce coherent combining:

$$\Phi_n(\Delta_n) = C + 2\pi m_n, \quad m_n \in \mathbb{Z}, \quad \forall n. \quad (13)$$

Equivalently,

$$\boxed{R_n(\Delta_n) + n_{\text{eff}}\Delta_n = C' + m_n\lambda}, \quad C' \triangleq \frac{C}{k_0}. \quad (14)$$

This is the PASS-specific “double optical-path” alignment condition.

3.3 Near-optimal aligned placement: target selection and solving

Fix the phase-alignment condition

$$R(\Delta) + n_{\text{eff}}\Delta = d_n, \quad d_n = C' + m_n\lambda, \quad m_n \in \mathbb{Z}, \quad (15)$$

where $R(\Delta) = \sqrt{d^2 + \Delta^2}$. We construct a near-optimal feasible placement by (i) selecting each target d_n via wavelength-grid snapping under spacing constraints and (ii) solving the corresponding one-dimensional equation for Δ .

Step 1: Feasible initialization and wavelength-grid snapping. Generate an initial right-half sequence satisfying the minimum spacing constraint:

$$\Delta_1^{(0)} = \frac{\Delta_p\lambda}{2}, \quad \Delta_{n+1}^{(0)} = \Delta_n^* + \Delta_p\lambda, \quad n = 1, \dots, N/2 - 1. \quad (16)$$

For each n , define the snapped target distance

$$d_n = \lambda \left\lceil \frac{1}{\lambda} \left(\sqrt{d^2 + (\Delta_n^{(0)})^2} + n_{\text{eff}}\Delta_n^{(0)} \right) \right\rceil. \quad (17)$$

Step 2: Solve for the aligned position (Newton / closed form). Given d_n , solve

$$\sqrt{d^2 + \Delta^2} + n_{\text{eff}}\Delta - d_n = 0. \quad (18)$$

Newton solver. Define $f(\Delta) = \sqrt{d^2 + \Delta^2} + n_{\text{eff}}\Delta - d_n$ with

$$f'(\Delta) = \frac{\Delta}{\sqrt{d^2 + \Delta^2}} + n_{\text{eff}}. \quad (19)$$

Iterate

$$\Delta^{(k+1)} = \Delta^{(k)} - \frac{f(\Delta^{(k)})}{f'(\Delta^{(k)})}, \quad (20)$$

initialized by $\Delta^{(0)} = \Delta_n^{(0)}$.

Closed-form solver (optional). Squaring (18) yields the quadratic equation

$$(n_{\text{eff}}^2 - 1)\Delta^2 - 2d_n n_{\text{eff}}\Delta + (d_n^2 - d^2) = 0, \quad (21)$$

hence for $n_{\text{eff}} \neq 1$,

$$\Delta = \frac{d_n n_{\text{eff}} \pm \sqrt{d_n^2 + d^2(n_{\text{eff}}^2 - 1)}}{n_{\text{eff}}^2 - 1}, \quad (22)$$

where the physically feasible root is selected to satisfy the side (right/left) and spacing constraints.

Step 3: Symmetry completion. Mirror the left-half positions by $\Delta_{-n}^* = -\Delta_n^*$.

Output. The constructed placement is $\mathbf{\Delta}^* = [\Delta_{-N/2}^*, \dots, \Delta_{-1}^*, \Delta_1^*, \dots, \Delta_{N/2}^*]^T$.

3.4 Baseline (ideal) array gain

Under perfect phase alignment, all phasors add constructively:

$$a_{\text{ideal}} = \frac{\eta}{N} \left(\sum_{n \in \mathcal{N}} \frac{1}{R_n(\Delta_n^*)} \right)^2. \quad (23)$$

For symmetric layouts with $|\Delta_n^*| \ll d$, $R_n(\Delta_n^*) \approx d$ and

$$a_{\text{ideal}} \approx \frac{\eta N}{d^2}. \quad (24)$$

4 Position Error Model and First-Order Perturbation

4.1 Bounded position errors

The actual offsets are perturbed as

$$\Delta_n = \Delta_n^* + \delta_n, \quad |\delta_n| \leq \varepsilon. \quad (25)$$

4.2 Phase sensitivity (first-order Taylor)

With $\Phi_n(\Delta) = k_0(\sqrt{d^2 + \Delta^2} + n_{\text{eff}}\Delta)$, a first-order expansion at Δ_n^* gives

$$\Phi_n(\Delta_n^* + \delta_n) \approx \Phi_n(\Delta_n^*) + \Phi_n'(\Delta_n^*)\delta_n, \quad (26)$$

where

$$\Phi_n'(\Delta) = k_0 \left(\frac{\Delta}{\sqrt{d^2 + \Delta^2}} + n_{\text{eff}} \right). \quad (27)$$

Define the composite sensitivity

$$\xi_n \triangleq \frac{\Delta_n^*}{\sqrt{d^2 + (\Delta_n^*)^2}} + n_{\text{eff}} = \sin \theta_n^* + n_{\text{eff}}, \quad (28)$$

so the phase error is approximately

$$\Delta\phi_n \triangleq \Phi_n(\Delta_n^* + \delta_n) - \Phi_n(\Delta_n^*) \approx k_0 \xi_n \delta_n. \quad (29)$$

The maximum phase deviation is

$$|\Delta\phi_n|_{\text{max}} \approx k_0 |\xi_n| \varepsilon. \quad (30)$$

Note that $\xi_n \in (n_{\text{eff}} - 1, n_{\text{eff}} + 1)$, hence $\xi_n > 0$ when $n_{\text{eff}} > 1$.

5 Worst-Case Array Gain at Δ^* : Analytical Bounds

5.1 Worst-case definition

We define the worst-case gain under bounded errors as

$$a_{\text{wc}}(\Delta^*, \varepsilon) \triangleq \min_{\|\delta\|_\infty \leq \varepsilon} a(\Delta^* + \delta). \quad (31)$$

5.2 A guaranteed lower bound via cosine projection

Let $w_n \triangleq 1/R_n(\Delta_n^*)$ and define

$$R_{n,\text{max}} \triangleq \sqrt{d^2 + (|\Delta_n^*| + \varepsilon)^2}, \quad w_n^- \triangleq \frac{1}{R_{n,\text{max}}}. \quad (32)$$

Assume $k_0 |\xi_n| \varepsilon \leq \pi/2$ for all n (small-error regime). Then each phasor $e^{-i\Delta\phi_n}$ has real part at least $\cos(k_0 |\xi_n| \varepsilon)$, which yields a deterministic projection bound:

$$a_{\text{wc}}(\Delta^*, \varepsilon) \geq \frac{\eta}{N} \left(\sum_{n \in \mathcal{N}} w_n^- \cos(k_0 |\xi_n| \varepsilon) \right)^2. \quad (33)$$

Simplified forms. If amplitude perturbations are ignored (common engineering approximation),

$$a_{\text{wc}}(\Delta^*, \varepsilon) \gtrsim \frac{\eta}{N} \left(\sum_{n \in \mathcal{N}} \frac{\cos(k_0 |\xi_n| \varepsilon)}{R_n(\Delta_n^*)} \right)^2. \quad (34)$$

Let $\xi_{\text{max}} \triangleq \max_n |\xi_n|$. In the same small-error regime,

$$a_{\text{wc}}(\Delta^*, \varepsilon) \geq a_{\text{ideal}} \cdot \cos^2(k_0 \xi_{\text{max}} \varepsilon) \geq a_{\text{ideal}} \cdot \cos^2\left(\frac{2\pi(n_{\text{eff}} + 1)\varepsilon}{\lambda}\right). \quad (35)$$

Remarks.

- The lower bound is placement-dependent through $\{R_n(\Delta_n^*)\}$ and $\{\xi_n\}$, thus enabling fair robustness comparison across different PASS placements (uniform symmetric, refinement-based, etc.).
- The term $k_0 \xi_n$ reveals scaling with frequency (k_0), waveguide index (n_{eff}), geometry (d), and aperture distribution ($\{\Delta_n^*\}$).

5.3 A constructive analytical upper bound on a_{wc}

An upper bound on the minimum can be obtained by exhibiting one feasible error pattern: for any δ with $\|\delta\|_\infty \leq \varepsilon$,

$$a_{\text{wc}}(\Delta^*, \varepsilon) = \min_{\|\delta\|_\infty \leq \varepsilon} a(\Delta^* + \delta) \leq a(\Delta^* + \delta_{\text{test}}). \quad (36)$$

A natural “phase-worst” corner pattern is $\delta_n = \varepsilon \cdot \text{sign}(\xi_n)$ (all phasors rotated in the same direction):

$$a_{\text{wc}}(\Delta^*, \varepsilon) \leq \frac{\eta}{N} \left| \sum_{n \in \mathcal{N}} \frac{1}{R_n(\Delta_n^* + \varepsilon \text{sign}(\xi_n))} \exp(-ik_0 \xi_n \varepsilon \text{sign}(\xi_n)) \right|^2. \quad (37)$$

This is computable in closed form once Δ^* is available (and can be tightened by testing multiple corner patterns, e.g., alternating $\pm\varepsilon$).

Remark (trivial envelope). Clearly $a_{\text{wc}}(\Delta^*, \varepsilon) \leq a_{\text{ideal}}$.

6 PASS vs. MA: Error Amplification Insight

For conventional MA (free-space-only) models, the phase sensitivity is dominated by $\partial R / \partial \Delta = \Delta / \sqrt{d^2 + \Delta^2} = \sin \theta \in (-1, 1)$. For PASS,

$$\xi_n = \sin \theta_n^* + n_{\text{eff}} \in (n_{\text{eff}} - 1, n_{\text{eff}} + 1), \quad (38)$$

so the waveguide term n_{eff} can dominate and amplify phase errors, making the array gain degrade faster with ε/λ .

7 Numerical Validation Plan

To validate the analytical bounds:

- 1) **Baseline construction:** Generate Δ^* via the refinement rule and solve each alignment equation by Newton (or the closed-form root).
- 2) **Worst-case search:** For each ε , compute

$$a_{\text{wc}}(\Delta^*, \varepsilon) = \min_{\|\delta\|_\infty \leq \varepsilon} a(\Delta^* + \delta), \quad (39)$$

numerically. Practical options:

- small N : enumerate corner points $\delta_n \in \{\pm\varepsilon\}$;
 - large N : coordinate descent / alternating minimization, where each δ_n is updated via a 1D search in $[-\varepsilon, \varepsilon]$ (often corner updates are sufficient).
- 3) **Bound comparison:** Plot a_{wc} versus ε together with the analytical lower bound and the constructive upper bound.

References (placeholders)

References

- [1] Array Gain for Pinching-Antenna Systems (PASS), arXiv preprint, 2025.
- [2] Movable Antenna-Enhanced Near-Field Beamforming: Impact of Antenna Position Errors, FCN 2025.